

QCA Fundamentals

Week 1: Classical to Quantum Transition

Cornell Quantum Computing Association

August 20, 2025

What We're Building Toward

- Classical computation $\rightarrow$ quantum algorithms on real hardware
- Mathematical tools: quantum gates
- Entanglement: quantum computing's real power
- Implement Deutsch's algorithm (quantum advantage)
- Quantum teleportation
- Major algorithms: Shor's, Grover's, quantum cryptography
- Real hardware: superconducting circuits, trapped ions
- Quantum error correction and practical computing

Why Quantum Computing Matters

- Not just faster - fundamentally different
- Exponential speedups for certain problems
- Applications: cryptography, drug discovery, optimization
- Major investments: paradigm shift recognition
- Current stage: noisy, error-prone, limited
- Timeline: 5-25 years depending on application

Course Format

- 5 minutes: review previous worksheet
- 55 minutes: interactive lecture with demonstrations
- Materials: slides, worksheets, Classiq/Qiskit notebooks
- No grades - focus on understanding
- Accessible regardless of background

Starting Simple: A Single Coin

Classical coin states:

State	Value	Vector
Heads	0	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
Tails	1	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Key insight: Completely in one state, completely not in the other

Multiple Coins: Exponential Growth

- 2 coins $\rightarrow$ 4 outcomes: (0,0), (0,1), (1,0), (1,1)
- 3 coins $\rightarrow$ 8 outcomes
- n coins $\rightarrow 2^n$ outcomes

Classical computing: n bits store one of 2^n values

Limitations of Classical Approach

To check all arrangements of n coins:

- Check $(0, 0, 0, \dots, 0)$
- Check $(0, 0, 0, \dots, 1)$
- Check $(0, 0, 1, \dots, 0)$
- $\vdots$
- Need 2^n total checks

300 coins = more checks than atoms in universe!

Quantum Notation and States

Classical to quantum transition:

Classical Vector	Quantum State
$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$ 0\rangle$
$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$ 1\rangle$

$|0\rangle$ and $|1\rangle$ form the **computational basis**

Introducing Superposition

Quantum difference: qubits can exist in superposition

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

In vector form:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

α, β are **probability amplitudes** (complex numbers)

Probabilities and Normalization

For state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

- Probability of measuring $|0\rangle$: $|\alpha|^2$
- Probability of measuring $|1\rangle$: $|\beta|^2$
- Normalization condition: $|\alpha|^2 + |\beta|^2 = 1$

Quantum Gates as Matrices

Quantum gates manipulate qubits as matrices:

Quantum NOT gate (X gate):

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Verification:

$$X |0\rangle = |1\rangle \tag{1}$$

$$X |1\rangle = |0\rangle \tag{2}$$

The Quantum Coin That Doesn't Land

- Quantum coin keeps "spinning" forever
- Not exactly heads, not exactly tails - both simultaneously
- Not classical uncertainty - genuine superposition
- No classical analog

Balanced superposition:

$$|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

Measuring the Quantum Coin

- Measurement forces coin to "choose"
- Superposition collapses to definite outcome
- From balanced superposition: 50% $|0\rangle$, 50% $|1\rangle$
- Before measurement: genuinely both states

Creating Superposition

Hadamard gate creates balanced superposition:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Action on $|0\rangle$:

$$H |0\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

Creates the perfectly balanced quantum coin!

Two Quantum Coins

Two quantum coins in superposition:

$$|\psi\rangle = \alpha_1 |00\rangle + \alpha_2 |01\rangle + \alpha_3 |10\rangle + \alpha_4 |11\rangle$$

Contains information about all 4 classical possibilities simultaneously

Normalization: $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2 = 1$

Scaling Up: The Exponential Advantage

Classical	Quantum
2^n computers	n qubits
Exponential resources	Logarithmic resources
1 million computers	20 qubits

The Catch: Measurement Collapse

- n qubits: exponentially many possibilities in superposition
- Measurement: collapse to just ONE classical outcome
- Can't directly read all 2^n pieces of information
- Get just one answer, chosen probabilistically

The Strategy: Interference

- Quantum amplitudes interfere like waves
- **Constructive interference:** amplify correct answers
- **Destructive interference:** cancel wrong answers
- Like tuning an instrument - right notes loud, wrong notes quiet

Simple Example: Finding a Pattern

Find arrangement of 8 coins with special property:

Classical: Check all 8 arrangements (worst case: 8 steps)

Quantum:

1. Put 8 qubits in superposition (all possibilities)
2. Design interference: correct answer constructive, wrong destructive
3. Measure once $\rightarrow$ right answer with high probability

8 steps $\rightarrow$ 1 quantum computation

Why Only Certain Problems Benefit

Quantum advantage requires:

1. Exponentially large search space
2. Ability to design interference patterns
3. Problem structure allowing quantum interference

Good for: factoring, searching, quantum simulation

No help: word processing, web browsing

What We Need

To build quantum algorithms:

1. Create superposition
2. Manipulate superposition
3. Create interference
4. Measure results

Next weeks: learn each tool systematically

Mathematical Tools We'll Develop

Next week:

- Bloch sphere visualization
- More quantum gates
- Born rule for measurement
- Multiple qubit systems
- Programming quantum circuits

From Theory to Algorithm

Coming up:

- Deutsch's algorithm: quantum advantage demo
- Quantum teleportation: impossible with classical
- Major algorithms: Shor's, Grover's
- Real hardware: superconducting circuits, trapped atoms

The Journey Ahead

Today's foundation:

- Classical coins $\rightarrow$ quantum coins
- Superposition and interference advantages
- Mathematical language for quantum computing

Goal: Harness quantum phenomena to solve impossible problems